Adyasha Maharana

8707 Braxwood Pl, Raleigh, NC 27617 | +1-253-355-7679 | adypooja@gmail.com

RESEARCH INTERESTS

Natural language processing: knowledge-augmented information extraction and inference
Biomedical image analysis: segmentation and interpretable prediction

EDUCATION

M.S. (Biomedical & Health Informatics)
University of Washington, Seattle, WA, 2015-2017

B.Tech. & M.Tech. (Integrated; Electronics & Electrical Communication Engineering with specialization in Visual Information & Embedded Systems)
Indian Institute of Technology, Kharagpur, WB, India, 2010-2015

PUBLICATIONS

- **Maharana, Adyasha**, and Elaine Okanyene Nsoesie. "Use of Deep Learning to Examine the Association of the Built Environment with Prevalence of Neighborhood Adult Obesity." JAMA Network Open 1, no. 4 (2018): e181535-e181535. [html]

- **Maharana, Adyasha**, and Meliha Yetisgen. "Clinical Event Detection with Hybrid Neural Architecture." BioNLP 2017 (2017): 351-355. [pdf]

- **Maharana, Adyasha**, Quynh C. Nguyen, and Elaine O. Nsoesie. "Using Deep Learning and Satellite Imagery to Quantify the Impact of the Built Environment on Neighborhood Crime Rates." arXiv preprint arXiv:1710.05483 (2017). [pdf]

- Nsoesie, Elaine O., Luisa Flor, Jared Hawkins, **Adyasha Maharana**, Tobi Skotnes, Fatima Marinho, and John S. Brownstein. "Social media as a sentinel for disease surveillance: what does sociodemographic status have to do with it?" PLoS currents 8 (2016). [html]

- Henly, Samuel, Gaurav Tuli, Sheryl A. Kluberg, Jared B. Hawkins, Quynh C. Nguyen, Aranka Anema, **Adyasha Maharana**, John S. Brownstein, and Elaine O. Nsoesie. "Disparities in digital reporting of illness: A demographic and socioeconomic assessment." Preventive medicine 101 (2017): 18-22. [pdf]

- **Maharana, Adyasha**, Arpit Tandon, Mihir Shah, Jason Philips, Ruchir Shah, and Brian Howard. " A Pragmatic Approach to Information Extraction for Systematic Review." (*to appear in TAC Proceedings 2018*)

PATENT

- Context Aware Delta Compression Algorithm for Genomic Files - A compression algorithm and storage method for intermediary files generated from genomic analysis pipelines (*US patent pending*)

THESIS

Extraction of clinical timeline from discharge summaries using neural networks

M.S., Advisor: Prof. Meliha Yetisgen, Biomedical & Health Informatics

- Clinical Entity Recognition: Developed a hybrid pipeline of Bi-LSTM-CRF, feature-based CRF to tag six classes of clinical entities and improve F1 score for rare events; Performed comparative analysis for out-of-domain performance of CRF, RNN systems on clinical notes
- Temporal Relation Extraction: Designed and implemented CNN-RNN architectures for extraction of temporal relations; Achieved 67% recall of within-sentence relations with bi-LSTM RNN using character, POS, entity class embeddings

Mixture-models based image segmentation algorithms: Applications in biomedical images

M.Tech., Advisor: Prof. Chandan Chakraborty, School of Medical Science & Technology

- Designed novel Gaussian-Rayleigh mixture model (GRMM) algorithm to improve detection of small exudates in retinal scans; Used GRMM with dynamic thresholding for unsupervised segmentation of regions of interest from brain MRIs

Modified Demon's algorithm for image registration and VLSI implementation

B.Tech., Advisor: Prof. Swapna Banerjee, Electronics & Electrical Communication Engineering

- Proposed and implemented an empirical update strategy for non-stationary weighting co-efficient in Modified Demon's algorithm resulting in 50% reduction in computation time, and thresholding of pixel selection resulting in 80% elimination of redundancy

WORK/RESEARCH EXPERIENCE

Data Scientist — Dec '17 – present

Sciome LLC, Research Triangle Park, NC

- Developed relation extraction pipeline for 18 entity classes and 3 relation classes from scientific articles for NIST's systematic review IE challenge; Models ranked 1st and 2nd in two tasks
- Explored precision-recall trade-off in multi-tasking architectures for low-resource NER
- Co-wrote a successful small business innovation grant (SBIR) to develop active-learning based information extraction platform for systematic reviews
- Designed and implemented CNN to classify high-resolution assay images with >98% accuracy, in collaboration with National Toxicology Program (NTP)
- Developed non-linear extrapolation methods for s1500+ gene expression values using autoencoders

Research Assistant — Mar '16 – Jun '16, Sep '16 – Jun '17

Institute of Health Metrics and Evaluation (IHME), Seattle, WA

- Designed and implemented study to gather urban neighborhood data from multiple sources (satellite imagery, places of interest, US census survey, 500 cities), extract relevant features using convolutional neural nets, transfer learning, elastic net regression and predict county-level estimates of obesity prevalence, annual crime rate
- Developed classification pipeline for highly-skewed dataset of crowd-sourced customer reviews to flag food products with high FDA recall potential

- Developed text classifier for classifying Portuguese tweets that represent actual cases of dengue; mapped tweet frequencies to actual dengue prevalence in three major states of Brazil

Research Intern — Jun '16 – Sep '16

IBM Almaden Research Center, San Jose, CA

- Genomic data compression: Conceived novel delta compression algorithm for storage of intermediary SAM, VCF files generated in genomic analysis pipeline, results in ~80% savings (*filed for patent*)
- Trace file parser: Developed Python script to automate parsing of Spectrum Scale traces, extract relevant info about I/O operations, compute statistics and present summary

Research Intern — Jun '14 – Aug '14

German Cancer Research Center (DKFZ), Heidelberg, Germany

- Shape from Shading: Laid groundwork for proposal to reconstruct 3D structure of colon from multiple colonoscopy 2D images. Proposed novel algorithm to inculcate non-linear information of specularities for reconstruction of colon

TEACHING EXPERIENCE

Teaching Assistant

Indian Institute of Technology, Kharagpur, India

Digital Signal Processing Theory and Laboratory — Jan '15 – Apr '15

Microcontrollers & Microprocessor Laboratory — Aug '14 – Nov '14

- Created and graded weekly assignments for students; collaborated with 4 other TA's to manage lab
- Provided hands-on assistance to students during lab sessions and held occasional lectures to explain concepts

GRANTS

PyLadies Travel Grant for presenting at PyCon — May '16, '17

Small Business Innovation Research (SBIR) Grant — Aug '18

CONFERENCE PRESENTATIONS

How to judge debates using Python (html), *PyCon* — May '16

Improving readability of online content by removing abusive speech using Python (html), *PyCon* — May '17

Mining online data for early identification of unsafe food products, *Data for Good Exchange* — Sep '18